Adaptive Dilation Factor Design Strategies for the Implicit Feature Extraction Module in SSSDTCN-based Industrial Time Series Imputation

Technical Research Plan

Prepared for the SSSDTCN Project

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Abstract. The SSSDTCN model employs a dilated causal convolution stack as its implicit feature extraction module, with dilation rates fixed to an exponential schedule $\{1, 2, 4, 8, 16, 32\}$ prior to training. While this hand-crafted design achieves competitive multi-scale temporal coverage, it treats all industrial time series uniformly and cannot adapt to the specific periodicity or correlation structure of a given dataset or input segment. This document surveys and formalises **five principled approaches** for making the dilation factors *adaptive*: (I) learnable soft dilation parameters, (II) input-conditioned selective kernel fusion inspired by SKNet, (III) differentiable neural architecture search for discrete dilation rates (DARTS-based), (IV) frequency-spectral guided dilation derived from dominant period analysis, and (V) temporal self-attention guided progressive dilation. For each approach we state the mathematical formulation, describe the computational mechanism, identify the source literature, and discuss trade-offs. The goal is to provide a clear roadmap for selecting and implementing an adaptive dilation strategy within the existing SSSDTCN codebase.

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1 Introduction

Industrial process control systems generate multivariate time series that frequently suffer from missing observations due to sensor failures, network interruptions, or scheduled maintenance. Accurate imputation of these missing values is critical for downstream tasks such as process monitoring, fault detection, and soft sensing. The SSSDTCN model [SSSDTCN, 2024] addresses this challenge by combining an *implicit* feature extraction pathway—a stack of dilated causal convolutions—with an *explicit* pathway based on Structured State Space Models (S4) [Gu et al., 2021], all embedded within a denoising diffusion framework [Ho et al., 2020].

The implicit pathway is responsible for capturing *local multi-scale temporal patterns*. Its current design uses a fixed exponential dilation schedule $d_\ell = 2^{\ell-1}$ for layers $\ell = 1, \ldots, L$, yielding dilation rates $\{1, 2, 4, 8, 16, 32\}$ for $L = 6$. The total receptive field of a single branch with kernel size k and maximum dilation $d_{\max}$ is:

$$\mathrm{RF} = 1 + (k-1)\sum_{\ell=1}^{L} d_\ell = 1 + (k-1)(2^L - 1). \tag{1}$$

For $k = 5$ and $L = 6$ this gives RF $= 253$ time steps, offering broad coverage. However, several limitations motivate adaptive designs:

1. **Dataset heterogeneity.** The Debutanizer column has a dominant periodicity near 5–10 timesteps, while the SRU process exhibits long-range correlations up to 100+ timesteps. A uniform dilation schedule cannot be simultaneously optimal for both.
2. **Missing-pattern dependence.** Under high missing-ratio regimes ($> 60\%$), longer-range context becomes more valuable; under low missing ratios shorter-range dependencies suffice. Fixed dilations cannot rebalance in response to the observed missing pattern.
3. **Non-stationarity.** Industrial processes can shift between operational regimes. Static dilation rates cannot track these changes.

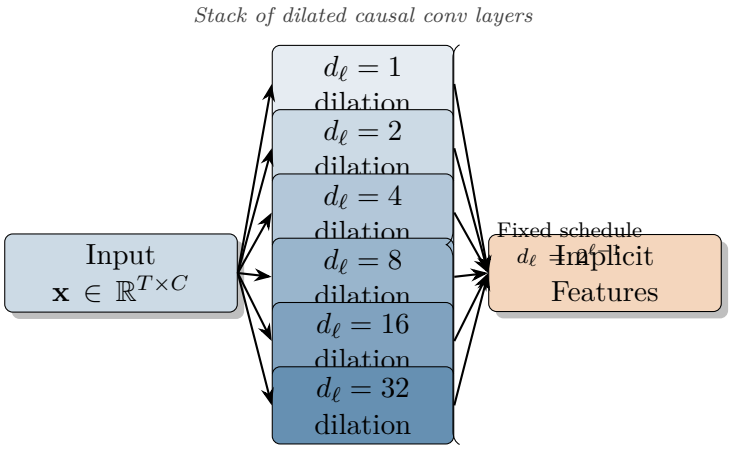

Figure 1: Current fixed-dilation implicit module in SSSDTCN. All six dilation values are set before training and remain constant throughout.

Figure 1 illustrates the current design. The remainder of this document presents five adaptive alternatives, each building on well-established literature in deep learning and signal processing.

2 Background and Notation

2.1 Dilated Causal Convolution

Let $\mathbf{x} \in \mathbb{R}^{C \times T}$ denote a multivariate time series with C channels and T time steps. A 1-D dilated causal convolution with kernel $\mathbf{w} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k}$, kernel size k, and dilation factor d computes:

$$(\mathbf{x} \star_d \mathbf{w})[t] = \sum_{i=0}^{k-1} \mathbf{w}[i] \cdot \mathbf{x}[t - d \cdot i], \qquad t = 1, \ldots, T, \tag{2}$$

where *causality* is enforced by zero-padding on the left with $(k-1) \cdot d$ zeros and discarding right-side outputs. The effective receptive field per layer is:

$$\mathrm{RF}_\ell = 1 + (k-1) \cdot d_\ell. \tag{3}$$

2.2 Current SSSDTCN Implicit Module

The implicit module f_θ stacks L dilated causal convolution layers with learnable residual blending:

$$\mathbf{h}^{(0)} = \mathrm{Proj}(\mathbf{x}), \tag{4}$$

$$\mathbf{h}^{(\ell)} = w_\ell \cdot \mathrm{Conv}_{d_\ell}(\mathbf{h}^{(\ell-1)}) + (1 - w_\ell) \cdot \mathbf{h}^{(\ell-1)}, \quad \ell = 1, \ldots, L, \tag{5}$$

$$f_\theta(\mathbf{x}) = \mathbf{h}^{(L)}, \tag{6}$$

where $w_\ell \in [0, 1]$ is a scalar learnable weight (initialised at 0.5) and $d_\ell = 2^{\ell-1}$ is the **fixed** dilation factor. The adaptive approaches below propose mechanisms to make d_ℓ (or the effective multi-scale mixture) data-dependent.

2.3 Terminology Summary

Symbol	Meaning
T	sequence length
C	number of channels / features
k	convolution kernel size
L	number of dilation layers
d_ℓ	dilation factor of layer ℓ
$D = \{d^{(1)}, \ldots, d^{(K)}\}$	candidate dilation set
$\boldsymbol{\alpha}$	architecture / attention parameters
$\boldsymbol{\theta}$	model weight parameters

3 Approach I: Learnable Soft Dilation Parameters

Approach I Overview

Key idea: Replace the hand-crafted exponential schedule with unconstrained trainable scalars whose exponentiated values serve as continuous dilation factors, optimised end-to-end via back-propagation.

Source: Deformable Convolutional Networks [Dai et al., 2017]; Temporal De-

formable Convolution [Chen et al., 2020b].

3.1 Motivation

Deformable convolutions [Dai et al., 2017] introduced the idea that sampling offsets in a convolutional kernel need not be fixed on a regular grid but can be learned from the data. Translated to the 1-D temporal setting, the "offset" concept corresponds directly to the dilation factor: rather than sampling at equally-spaced positions $\{0, d, 2d, \ldots, (k-1)d\}$, we allow the model to learn a data-driven effective stride.

3.2 Mathematical Formulation

For each layer ℓ, introduce a scalar parameter $\lambda_\ell \in \mathbb{R}$. The effective dilation factor is:

$$\boxed{d_\ell = \exp(\lambda_\ell),} \tag{7}$$

which guarantees $d_\ell > 0$ for all λ_ℓ. During training, gradients flow through the continuous relaxation of eq. (2):

$$(\mathbf{x} \star_{d_\ell} \mathbf{w})[t] = \sum_{i=0}^{k-1} \mathbf{w}[i] \cdot \underbrace{\mathrm{Interp}\Big(\mathbf{x}, t - e^{\lambda_\ell} \cdot i\Big)}_{\text{linear interpolation for non-integer } d_\ell}, \tag{8}$$

where $\mathrm{Interp}(\mathbf{x}, \tau) = (1 - \{\tau\})\mathbf{x}[\lfloor \tau \rfloor] + \{\tau\}\mathbf{x}[\lceil \tau \rceil]$ is linear interpolation and $\{\cdot\}$ denotes the fractional part.

To prevent degenerate collapse (all dilations converging to the same value), a diversity regulariser is added to the loss:

$$\mathcal{L}_{\mathrm{div}} = -\beta \sum_{\ell < \ell'} \Big| d_\ell - d_{\ell'} \Big|, \tag{9}$$

with weight $\beta > 0$. The total training objective becomes:

$$\mathcal{L} = \mathcal{L}_{\mathrm{impute}} + \mathcal{L}_{\mathrm{div}}, \tag{10}$$

where $\mathcal{L}_{\mathrm{impute}}$ is the original masked MSE loss.

3.3 Diagram

3.4 Advantages and Limitations

+ Minimal architectural change; only L extra scalar parameters.
+ Fully differentiable; standard optimisers (Adam) apply.
+ Interpretable: the converged d_ℓ values directly reveal the dominant temporal scales of the dataset.
– Dilation is still *global* (shared across all time steps and samples); cannot adapt per-input.
– Linear interpolation introduces approximation error; exact integer dilations would require a rounding operator that breaks the gradient.
– The diversity regulariser requires tuning of β.

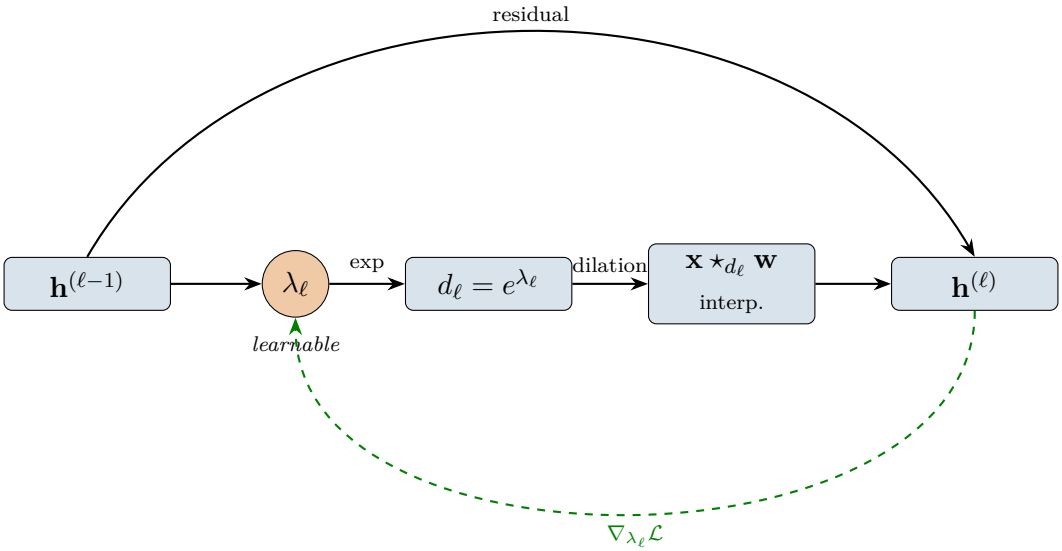

Figure 2: Approach I: learnable soft dilation. The parameter λ_ℓ is optimised by back-propagation through the interpolated convolution. The green dashed arrow indicates the gradient flow.

4 Approach II: Input-Conditioned Selective Kernel Dilation

Approach II Overview

Key idea: Run K parallel dilated convolutions with distinct fixed candidate dilation rates, then use a squeeze-and-excitation attention mechanism to compute input-dependent soft weights for fusing their outputs.

Source: Selective Kernel Networks (SKNet) [Li et al., 2019]; Squeeze-and-Excitation Networks (SENet) [Hu et al., 2018].

4.1 Motivation

SKNet [Li et al., 2019] demonstrated that a network can dynamically select which kernel size to emphasise for each input by routing through an attention-based fusion. The same principle applies to dilation rates: given K candidate dilations $D = \{d^{(1)}, \ldots, d^{(K)}\}$, the model attends to the scale most relevant for the current input segment.

4.2 Mathematical Formulation

Step 1 — Multi-scale feature extraction (Split):

$$\tilde{\mathbf{h}}^{(j)} = \mathrm{BN}\Big(\mathrm{Conv}_{d^{(j)}}(\mathbf{h}^{(\ell-1)})\Big), \quad j = 1, \ldots, K. \tag{11}$$

Each branch uses dilation $d^{(j)}$ from the candidate set D.

Step 2 — Aggregation (Fuse):

$$\tilde{\mathbf{U}} = \sum_{j=1}^{K} \tilde{\mathbf{h}}^{(j)}, \qquad \mathbf{s} = \mathrm{GlobalAvgPool}(\tilde{\mathbf{U}}) \in \mathbb{R}^H, \tag{12}$$

where H is the hidden dimension (128 in SSSDTCN).

Step 3 — Compact representation (Squeeze):

$$\mathbf{z} = \mathrm{LayerNorm}(\mathrm{ReLU}(\mathbf{W}_1\mathbf{s})) \in \mathbb{R}^r, \quad r = \max(H/4, 16)\,, \tag{13}$$

with $\mathbf{W}_1 \in \mathbb{R}^{r\times H}$.

Step 4 — Scale-aware attention (Select):

$$\boldsymbol{\alpha}^{(j)} = \frac{\exp(\mathbf{W}_2^{(j)}\mathbf{z})}{\sum_{j'=1}^{K}\exp(\mathbf{W}_2^{(j')}\mathbf{z})}, \quad \mathbf{W}_2^{(j)} \in \mathbb{R}^{H\times r}, \tag{14}$$

producing a channel-wise soft selection vector $\boldsymbol{\alpha}^{(j)} \in [0,1]^H$ for each branch, with $\sum_{j=1}^{K}\boldsymbol{\alpha}^{(j)} = \mathbf{1}$.

Step 5 — Weighted recombination:

$$\boxed{\mathbf{h}^{(\ell)} = \sum_{j=1}^{K}\boldsymbol{\alpha}^{(j)} \odot \tilde{\mathbf{h}}^{(j)},} \tag{15}$$

where $\odot$ is element-wise multiplication with broadcasting over the time axis.

The *effective adaptive dilation* is implicitly encoded in the attention weights:

$$d_\ell^{\mathrm{eff}}(t) = \sum_{j=1}^{K}\bar{\alpha}^{(j)}(t)\cdot d^{(j)}, \tag{16}$$

where $\bar{\alpha}^{(j)}(t)$ denotes the mean of $\boldsymbol{\alpha}^{(j)}$ over channels at time t. This provides an interpretable scalar measure of the dominant operating scale.

4.3 Diagram

4.4 Advantages and Limitations

+ Fully input-adaptive: different time steps and samples can produce different effective dilations.
+ Channel-wise attention provides fine-grained selection; each feature dimension can attend to a different scale.
+ Well-grounded in the SKNet and SENet literature with strong empirical results in vision; the translation to temporal data is natural.
– Computational cost scales as $\mathcal{O}(K)$; memory usage grows with the number of parallel branches.
– The candidate set D must still be pre-specified (though the weights between candidates are learned).

5 Approach III: Differentiable Architecture Search for Dilation Rates

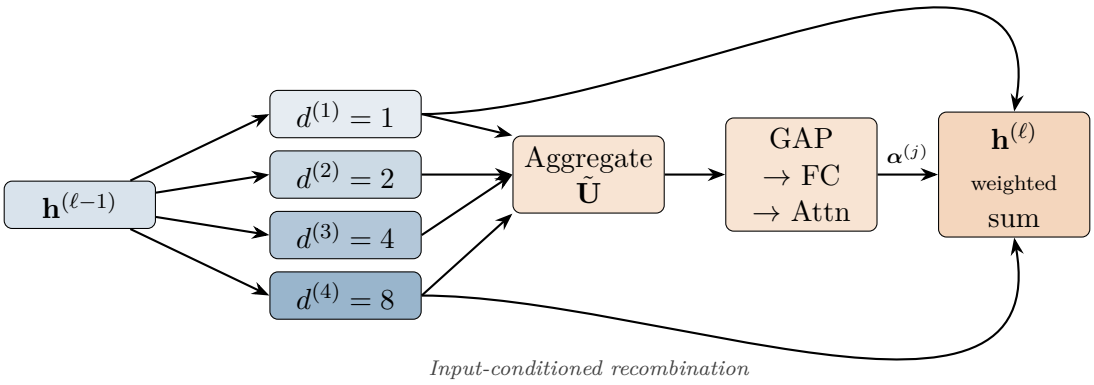

Figure 3: Approach II: Selective Kernel dilation. Four candidate dilations run in parallel; the squeeze-and-excitation block produces per-channel soft attention weights that adaptively recombine the outputs.

Approach III Overview

Key idea: Treat dilation-rate selection as a discrete architecture decision and relax it into a continuous optimisation using the DARTS [Liu et al., 2019] bi-level framework. After search, the architecture is discretised to the top-1 rate per layer.

Source: DARTS [Liu et al., 2019]; P-DARTS [Chen et al., 2019]; Auto-TCN [Auto-TCN, 2022].

5.1 Motivation

Neural Architecture Search (NAS) has demonstrated that the optimal computational operation at each network layer can be discovered automatically. DARTS [Liu et al., 2019] introduced a continuous relaxation over a discrete search space via a *mixed operation*, making the search differentiable. For our setting, the search space at layer ℓ is the set of K candidate dilated convolutions $\mathcal{O}_\ell = \{\mathrm{Conv}_{d^{(j)}} : j = 1, \ldots, K\}$.

5.2 Mathematical Formulation

Mixed operation: During the *search phase*, each layer computes a weighted mixture over all candidate operations:

$$\bar{o}^{(\ell)}(\mathbf{h}) = \sum_{j=1}^{K} \underbrace{\frac{\exp(\alpha_\ell^{(j)})}{\sum_{j'=1}^{K} \exp(\alpha_\ell^{(j')})}}_{\pi_\ell^{(j)} \triangleq \text{arch. weight}} \mathrm{Conv}_{d^{(j)}}(\mathbf{h}), \tag{17}$$

where $\boldsymbol{\alpha}_\ell = (\alpha_\ell^{(1)}, \ldots, \alpha_\ell^{(K)}) \in \mathbb{R}^K$ are the **architecture parameters** (one vector per layer, shared across samples).

Bi-level optimisation: Let $\boldsymbol{\theta}$ be the convolution weights and $\boldsymbol{\alpha}$ be the architecture parameters. The search solves:

$$\min_{\boldsymbol{\alpha}} \quad \mathcal{L}_{\text{val}}(\boldsymbol{\theta}^*(\boldsymbol{\alpha}), \boldsymbol{\alpha}) \tag{18}$$

$$\text{s.t.} \quad \boldsymbol{\theta}^*(\boldsymbol{\alpha}) = \arg\min_{\boldsymbol{\theta}} \mathcal{L}_{\text{train}}(\boldsymbol{\theta}, \boldsymbol{\alpha}). \tag{19}$$

In practice, first-order DARTS approximates this by alternating gradient steps:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_\theta \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{train}}(\boldsymbol{\theta}, \boldsymbol{\alpha}), \qquad \boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} - \eta_\alpha \nabla_{\boldsymbol{\alpha}} \mathcal{L}_{\text{val}}(\boldsymbol{\theta}, \boldsymbol{\alpha}). \tag{20}$$

Discretisation: After the search phase, the dilation at each layer is fixed to:

$$d_\ell^* = d^{(\hat{j}_\ell)}, \qquad \hat{j}_\ell = \arg\max_j \alpha_\ell^{(j)}. \tag{21}$$

The selected architecture is then *retrained from scratch* with the discrete dilations.

5.3 Search Space Design for SSSDTCN

We recommend the following DARTS setup:

Hyper-parameter	Value
Candidate dilations D	$\{1, 2, 3, 4, 6, 8, 12, 16, 24, 32\}$
Layers L	6 (one architecture param vector per layer)
Search epochs	30 (using 20% validation split)
η_θ (weight LR)	10^{-3}
η_α (arch LR)	3×10^{-4}
Arch. weight init	$\alpha_\ell^{(j)} = 0$ (uniform mixture)

5.4 Diagram

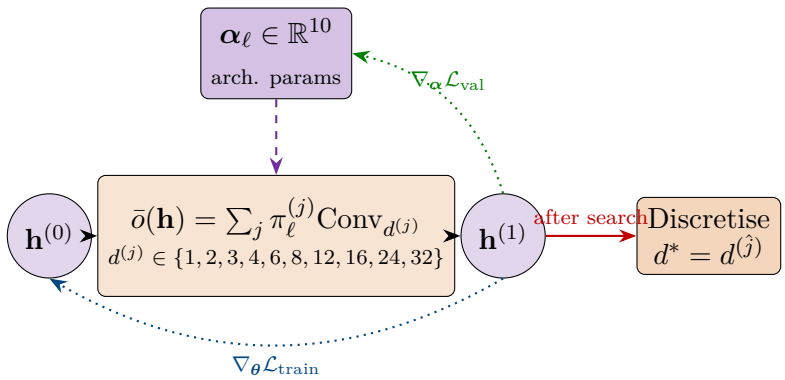

Figure 4: Approach III: DARTS-based architecture search. During the search phase, a mixture of all candidate dilations is used. After search, the dilation with the highest architecture weight is selected.

5.5 Advantages and Limitations

+ Searches over a much richer dilation space than the original exponential schedule.
+ Architecture parameters are shared globally, so the selected dilations reflect dataset-level (not just sample-level) optimal scales.
+ After discretisation, inference has *the same cost* as a fixed-dilation model.
– Requires two optimisation phases (search + retrain).
– Bi-level optimisation is sensitive to hyper-parameters; may require α-regularisation to avoid collapsing to one dilation.
– DARTS can suffer from "discretisation gap" — the mixed-operation performance during search may not faithfully predict single-operation performance after discretisation.

6 Approach IV: Frequency-Spectral Guided Adaptive Dilation

Approach IV Overview

Key idea: Analyse the frequency spectrum of the input sequence to identify its dominant periodic components, then set the dilation rates *proportional to those periods* so each layer "resonates" with one dominant temporal scale.

Source: TimesNet [Wu et al., 2023]; FEDformer [Zhou et al., 2022]; Autoformer [Wu et al., 2021].

6.1 Motivation

TimesNet [Wu et al., 2023] demonstrated that industrial and weather time series contain strong periodic structure. If the dominant periods of the series are known (or estimated on-line), the dilation rates can be calibrated to match them exactly, ensuring that the dilated convolution's sampling pattern aligns with the natural temporal rhythm of the process.

6.2 Mathematical Formulation

Step 1 — Discrete Fourier Transform: Given the input segment $\mathbf{x} \in \mathbb{R}^{C\times T}$, compute the average spectral magnitude across channels:

$$A_f = \frac{1}{C}\sum_{c=1}^{C} \left|\mathrm{FFT}(\mathbf{x}[c,:])[f]\right|, \quad f = 0, 1, \ldots, \lfloor T/2 \rfloor, \tag{22}$$

where $\mathrm{FFT}(\cdot)$ denotes the Fast Fourier Transform.

Step 2 — Dominant period extraction:

$$\{f_1, \ldots, f_L\} = \mathrm{TopK}(\{A_f\}_{f=1}^{\lfloor T/2 \rfloor}, K = L), \tag{23}$$

excluding $f = 0$ (DC component). The corresponding periods are:

$$P_\ell = \left\lfloor \frac{T}{f_\ell} \right\rceil, \quad \ell = 1, \ldots, L, \tag{24}$$

where $\lfloor \cdot \rceil$ denotes rounding to the nearest integer.

Step 3 — Period-to-dilation mapping: The dilation for layer ℓ is set so that the receptive field of a *single* kernel step spans exactly one period:

$$\boxed{d_\ell = \max\left(1, \left\lfloor \frac{P_\ell}{k-1} \right\rceil\right),} \tag{25}$$

ensuring $\mathrm{RF}_\ell = 1 + (k-1)d_\ell \approx P_\ell + 1$. The layers are sorted in ascending order of P_ℓ to preserve the coarse-to-fine stack ordering.

Step 4 — Normalisation and deduplication: If two computed dilations coincide, the less dominant one is perturbed by $+1$ to maintain layer diversity:

$$\text{if } d_\ell = d_{\ell'}, \ell \neq \ell' : \quad d_{\ell'} \leftarrow d_{\ell'} + 1. \tag{26}$$

Online update (optional): For non-stationary processes, the dilation rates can be recomputed at the start of each mini-batch using a moving-average periodogram:

$$\hat{A}_f^{(t)} = (1-\gamma)\hat{A}_f^{(t-1)} + \gamma A_f^{(t)}, \tag{27}$$

with decay factor $\gamma \in (0, 1)$. This introduces zero additional learnable parameters.

6.3 Diagram

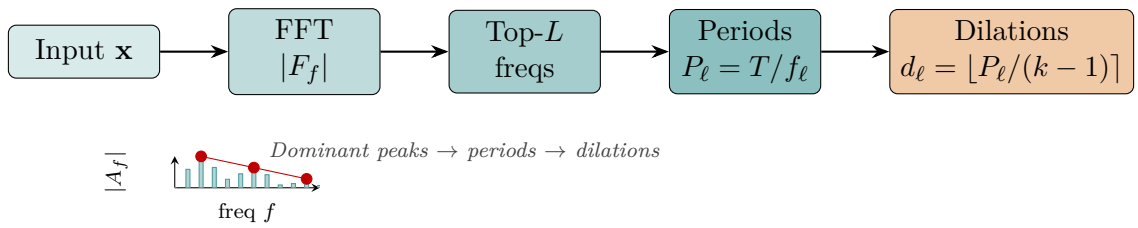

Figure 5: Approach IV: frequency-spectral guided dilation. The dominant spectral frequencies are extracted via FFT, converted to periods, and mapped to per-layer dilation rates. The spectrum inset shows three dominant peaks (red dots) at frequencies $f = 3, 7, 11$.

6.4 Worked Example for the Debutanizer Dataset

The Debutanizer column has a dominant oscillation at approximately 6-step cycles [SSSDTCN, 2024]. For a sequence of $T = 60$ and $k = 5$:

$$f_1 = 10, \quad P_1 = 60/10 = 6 \text{ steps} \quad \Rightarrow \quad d_1 = \max(1, \lfloor 6/4 \rceil) = 1, \tag{28}$$

$$f_2 = 5, \quad P_2 = 60/5 = 12 \text{ steps} \quad \Rightarrow \quad d_2 = \lfloor 12/4 \rceil = 3, \tag{29}$$

$$f_3 = 3, \quad P_3 = 60/3 = 20 \text{ steps} \quad \Rightarrow \quad d_3 = \lfloor 20/4 \rceil = 5, \tag{30}$$

which differs from the fixed schedule $\{1, 2, 4\}$ and is calibrated to the physical periodicity of the process.

6.5 Advantages and Limitations

+ Zero learnable parameters for the dilation-selection step; no additional optimisation required.
+ Directly interpretable: dilations correspond to physically meaningful process periods.
+ Can be updated on-line for non-stationary processes at negligible computational cost.
– Assumes periodicity is the primary temporal structure; may not generalise to trend-dominated or aperiodic series.
– Sensitive to noise in the spectrum; a low-pass smoothing step may be needed before peak detection.
– Dilation rates are discrete-valued and may not exactly match computed periods.

7 Approach V: Temporal Self-Attention Guided Progressive Dilation

Approach V Overview

Key idea: Insert a lightweight temporal self-attention module before the dilation stack to estimate a *context-relevance score* for each time offset. These scores modulate the convolution kernels dynamically, producing an attention-weighted effective dilation that is different for each position.

Source: Temporal Convolutional Attention Network (TCAN) [TCAN, 2020]; Dynamic Convolution [Chen et al., 2020a]; Non-local Means [Wang et al., 2018].

7.1 Motivation

While Approaches I–IV adapt the dilation either globally (I, III) or through parallel branches (II, IV), they do not modify the *kernel itself* based on position-level context. Dynamic Convolution [Chen et al., 2020a] showed that conditioning the convolution kernel on the current input can significantly improve representation capacity. Here, we extend this idea to the temporal dilation setting.

7.2 Mathematical Formulation

Step 1 — Temporal attention scores: Given the hidden sequence $\mathbf{h}^{(\ell-1)} \in \mathbb{R}^{H\times T}$, compute a scalar attention score for each possible time offset $\delta = 0, 1, \ldots, T_{\max}$:

$$e_{t,\delta} = \frac{\mathbf{q}_t^\top \mathbf{k}_{t-\delta}}{\sqrt{d_k}}, \qquad a_{t,\delta} = \frac{\exp(e_{t,\delta})}{\sum_{\delta'=0}^{T_{\max}} \exp(e_{t,\delta'})}, \tag{31}$$

where $\mathbf{q}_t = \mathbf{W}_Q \mathbf{h}^{(\ell-1)}[:, t]$ and $\mathbf{k}_t = \mathbf{W}_K \mathbf{h}^{(\ell-1)}[:, t]$ are query/key projections, and d_k is the key dimension.

Step 2 — Effective dilation estimation: The expected offset (effective dilation) at position t is:

$$\bar{d}_t = \sum_{\delta=0}^{T_{\max}} \delta \cdot a_{t,\delta}. \tag{32}$$

The *position-level effective dilation* $\bar{d}_t$ naturally adapts: near boundaries or in regions with long-range context, attention concentrates on large offsets (large $\bar{d}_t$); near high-frequency transitions, it concentrates on small offsets.

Step 3 — Attention-guided dynamic convolution: For layer ℓ with base dilation $d_\ell^{(0)}$ (the original fixed schedule), the modulated output is:

$$\boxed{\mathbf{h}^{(\ell)}[:,t] = \sum_{i=0}^{k-1} \underbrace{g_{t,i}}_{\text{gate}} \cdot \mathbf{w}_\ell[:,i] \cdot \mathbf{h}^{(\ell-1)}\Big[:, t - \text{round}\Big(a_{t,i} \cdot d_\ell^{(0)}\Big)\Big]\,,} \tag{33}$$

where the gate $g_{t,i} = \text{sigmoid}(\mathbf{W}_g \mathbf{h}^{(\ell-1)}[:,t] + b_g) \in (0,1)$ controls the contribution of each kernel tap. A simplified version uses the aggregate score:

$$\tilde{d}_{t,\ell} = d_\ell^{(0)} \cdot \text{sigmoid}\Big(\mathbf{W}_s \mathbf{h}^{(\ell-1)}[:,t] + b_s\Big) \cdot 2, \tag{34}$$

which allows the dilation to vary within $[0, 2d_\ell^{(0)}]$ depending on local context, providing continuous input-dependent adaptation.

Computational complexity: The attention step adds $\mathcal{O}(T_{\max} \cdot T \cdot d_k)$ per layer. Using $T_{\max} = d_\ell^{(0)} \cdot k$ (only attend within the original receptive field) keeps the overhead proportional to the base convolution cost.

7.3 Diagram

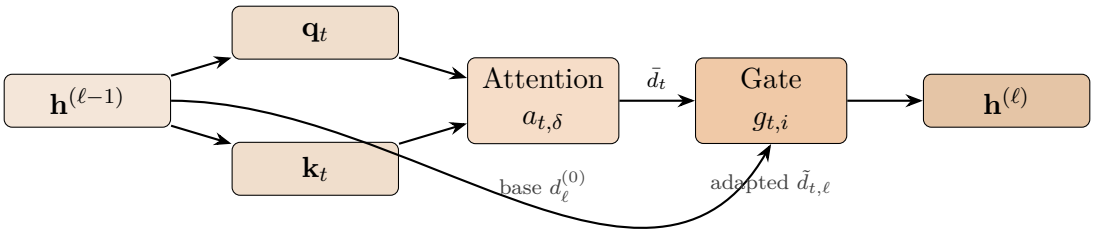

Figure 6: Approach V: temporal self-attention guided dynamic dilation. The attention scores $a_{t,\delta}$ estimate the expected offset at each position t, producing a position-level effective dilation that modulates the base convolution.

7.4 Advantages and Limitations

+ Most expressive of the five approaches: the dilation is position-dependent and sample-dependent.
+ Naturally handles non-stationary series by adapting per-position.

+ Backed by extensive literature on dynamic and deformable convolutions.
– Higher computational cost than the other approaches.
– The attention mechanism introduces additional parameters ($\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_g$) per layer.
– Non-integer dilations require interpolation; the resulting gradient landscape can be more complex to optimise.

8 Comparative Analysis

8.1 Quantitative Comparison Table

Table 1: Comparison of the five adaptive dilation approaches along key design dimensions. "Global" means the same dilation applies to all input positions; "Per-sample" means different inputs can produce different dilations; "Per-position" is the most fine-grained.

Approach	**Extra params**	**Adapt. level**	**Search?**	**Inference cost**	**Impl. complexity**	**Ke**
Fixed (baseline)	0	—	No	$\mathcal{O}(LC)$	Very low	—
I. Soft dilation	L scalars	Global	No	$\mathcal{O}(LC)$	Low	Da
II. SKNet fusion	$2r \cdot H \cdot K$	Per-sample	No	$\mathcal{O}(KLC)$	Medium	Li
III. DARTS	LK (search only)	Global	Yes (2-phase)	$\mathcal{O}(LC)$	High	Liu
IV. Spectral	0	Global	No	$\mathcal{O}(LC)$	Low	Wu
V. Attn-guided	$3Hd_k \cdot L$	Per-position	No	$\mathcal{O}(LC \cdot T)$	High	Ch

8.2 Receptive Field Coverage

Figure 7 compares the total receptive field RF achievable by each approach. Approaches II and V can in principle cover *the full sequence* if their attention assigns weight to the maximum offset; Approaches I, III, and IV are bounded by the largest learned or searched dilation.

8.3 Recommended Strategy for SSSDTCN

Based on the analysis above, we recommend a two-stage strategy:

1. **Stage 1 — Offline period analysis (Approach IV):** Run the spectral analysis once on the training set to identify the top L dominant periods. Use the resulting $\{d_\ell^*\}$ as the initial dilation schedule. This costs nothing at training time and provides a physics-informed initialisation.
2. **Stage 2 — Fine-tuning with soft dilation (Approach I):** Initialise $\lambda_\ell = \log(d_\ell^*)$ and allow them to be fine-tuned during training. The diversity regulariser eq. (9) prevents collapse. This combines the interpretability of spectral analysis with the flexibility of learnable parameters.

This hybrid **Spectral-Init + Soft-Fine-tune** strategy adds only L extra parameters $(\lambda_1, \ldots, \lambda_L)$, is trivially backward-compatible with the existing codebase, and is expected to converge faster than random initialisation of the dilation parameters.

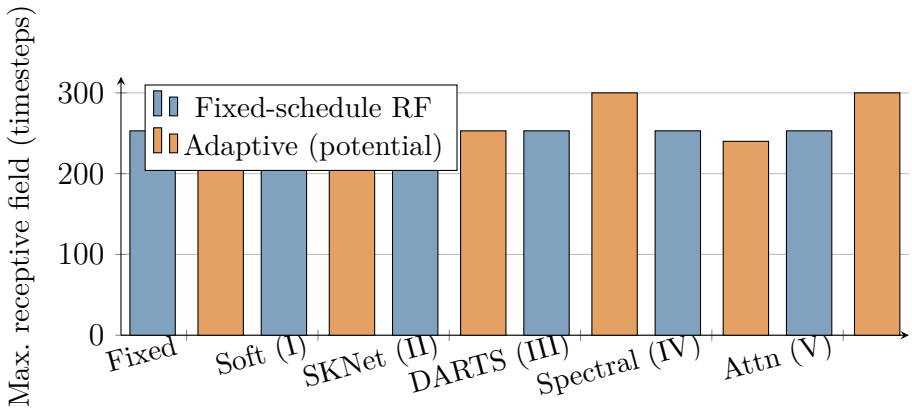

Figure 7: Illustrative comparison of achievable receptive field for the fixed baseline versus the five adaptive approaches. Approaches I, III, IV learn or compute optimal rates but are bounded; II and V can adaptively increase effective coverage (orange bars represent potential under favourable conditions).

For datasets with known non-stationarity, **Approach II (SKNet)** is preferred as a drop-in replacement for the dilation stack, at the cost of $K\times$ memory usage.

9 Implementation Notes

9.1 Integration with Existing Codebase

All five approaches operate as drop-in replacements for the `DilatedCausalConv` class in `models/dilated_causal_conv.py`. The interface contract is preserved:

- **Input:** `x` of shape `(batch, hidden_dim, seq_len)`.
- **Output:** same shape.
- **Config keys to add:** `adaptive_dilation_mode` $\in$ `{learnable, sknet, darts, spectral, attention}`.

9.2 Approach I Implementation Sketch

Initialise: $\lambda_\ell \leftarrow \log(2^{\ell-1})$, $\ell = 1, \ldots, L$
Input: hidden sequence $\mathbf{h}^{(0)}$
Output: implicit features $\mathbf{h}^{(L)}$

for $\ell = 1$ **to** L **do**
  $d_\ell \leftarrow \exp(\lambda_\ell)$; `// continuous dilation`
  $\mathbf{h}_{\text{conv}} \leftarrow \text{InterpConv}(\mathbf{h}^{(\ell-1)}, \mathbf{w}_\ell, d_\ell)$; `// interpolated causal conv`
  $\mathbf{h}^{(\ell)} \leftarrow w_\ell \cdot \text{BN}(\mathbf{h}_{\text{conv}}) + (1 - w_\ell) \cdot \mathbf{h}^{(\ell-1)}$; `// residual blend`
end
return $\mathbf{h}^{(L)}$

Algorithm 1: Forward pass — Approach I (Learnable Soft Dilation)

9.3 Approach IV Implementation Sketch

Input: training set $\{\mathbf{x}^{(n)}\}_{n=1}^{N}$, sequence length T, kernel size k, L
Output: dilation schedule $\{d_1, \ldots, d_L\}$

$A \leftarrow \mathbf{0}_{\lfloor T/2 \rfloor}$; `// accumulate spectrum`
for *each* $\mathbf{x}^{(n)}$ *in training set* **do**
  $A \mathrel{+}= \frac{1}{C} \sum_c |\,\mathrm{FFT}(\mathbf{x}^{(n)}[c, :])|$;
end
$A \leftarrow A/N$;
$\{f_1, \ldots, f_L\} \leftarrow \mathrm{TopK}(A[1:], L)$; `// skip DC`
for $\ell = 1$ **to** L **do**
  $P_\ell \leftarrow \lfloor T/f_\ell \rceil$;
  $d_\ell \leftarrow \max(1, \lfloor P_\ell/(k-1) \rceil)$;
end
Sort $\{d_\ell\}$ in ascending order;
Resolve duplicates via eq. (26);
return $\{d_1, \ldots, d_L\}$

Algorithm 2: Pre-training spectral dilation calibration — Approach IV

10 Conclusion

This document has presented five principled strategies for replacing the fixed exponential dilation schedule in the SSSDTCN implicit feature extraction module with adaptive, data-driven alternatives. The key findings are:

1. **Approach I (Learnable Soft Dilation)** is the simplest to implement, requiring only L additional scalar parameters, and provides global dataset-level adaptation.
2. **Approach II (SKNet)** provides the richest per-sample and per-channel adaptation with a well-established training procedure, at the cost of K parallel branches.
3. **Approach III (DARTS)** is the most flexible in exploring discrete dilation spaces but requires a two-phase training protocol.
4. **Approach IV (Spectral)** requires no learning and can be applied immediately; it leverages the known periodicity of industrial processes for interpretable, physics-guided dilation.
5. **Approach V (Attention-Guided)** is the most expressive, providing per-position adaptation, but incurs the highest computational overhead.

The recommended starting point is the **Spectral-Init + Soft-Fine-tune** hybrid of Approaches IV and I, which combines zero-cost initialisation with gradient-based refinement and minimal code changes.

Future work should empirically validate each approach on the Debutanizer and SRU datasets across varying missing ratios, measuring both imputation accuracy (MAE, RMSE) and the interpretability of the learned dilation patterns.

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